

Ji Young Kim

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RESEARCH INTERESTS AI Agents, Reasoning Trajectories, Failure Prediction, Program Synthesis & Automated Debugging, Retrieval-Augmented Generation (RAG), Software Engineering for AI Systems

EDUCATION **University of Illinois Urbana-Champaign** Aug. 2026-Present
Ph.D. in Computer Science and Engineering, Advisor: Prof. Reyhaneh Jabbarvand

Seoul National University Mar. 2015-Feb. 2017
Master of Science in Computer Science and Engineering

Korea University Mar. 2010-Aug. 2014
Bachelor of Science in Computer and Communication Engineering

- PUBLICATIONS
- [1] **J. Kim**, J. Dragovic, A. Botta, T. M. R. Islam, A. Mohamad, K. Sharaf, S. S. Siddiqui, D. Joshi, H. Anand, N. Saryyeva, S. Chapagain, S. Nasir, D. Marinov, and B. A. Stoica, "The Discreet Charm of the Bugeoisie: A First Look at Bug Reports Created by Researchers," in *Proc. ACM International Symposium on Software Testing and Analysis (ISSTA '26)*, Oakland, CA, USA, 2026, to appear.
 - [2] S. Bae, D. Ban, D. Han, **J. Kim**, K.-h. Lee, S. Lim, W. Park, and C.-k. Kim, "StreetSense: Effect of bus Wi-Fi APs on pedestrian smartphone," in *Proc. 2015 Internet Measurement Conf. (IMC '15)*, New York, NY, USA, 2015, pp. 347–353, doi: [10.1145/2815675.2815715](https://doi.org/10.1145/2815675.2815715)
 - [3] J. Park, **J. Kim**, K. Kim, S. Jang, and C.-k. Kim, "CTI in Wi-Fi/Bluetooth combo modules," poster presented at the *13th USENIX Symp. Networked Systems Design and Implementation (NSDI '16)*, Santa Clara, CA, USA, Mar. 2016
 - [4] J. Chung, D. Han, **J. Kim**, and C.-k. Kim, "Machine learning based path management for mobile devices over MPTCP," in *Proc. 2017 IEEE Int. Conf. Big Data Smart Comput. (BigComp)*, Jeju, Korea (South), 2017, pp. 206–209, doi: [10.1109/BIGCOMP.2017.7881739](https://doi.org/10.1109/BIGCOMP.2017.7881739)
 - [5] **J. Kim**, D. Han, and C.-k. Kim, "Impact of diverse document-evaluation measure-based searching methods in big data search accuracy," *J. KIISE*, vol. 44, no. 5, pp. 553–558, 2017.

RESEARCH EXPERIENCE **University of Illinois Urbana-Champaign (UIUC)**
Graduate Research Assistant | Advisor: Prof. Reyhaneh Jabbarvand Aug. 2026-Present

Process-level analysis of LLM-based software engineering agents to understand, diagnose, and improve behavior.

- **Agent Trajectory Analysis & Failure Diagnosis** Analyzing reasoning and action trajectories of coding agents to characterize behavioral patterns and identify early indicators of success and failure.
- **Process-Level Agent Evaluation** Developing metrics and empirical methods to study agent planning, exploration, convergence, and failure patterns beyond final task outcomes.
- **Agent Learning & Optimization** Investigating process-level supervision and reward signals derived from agent trajectories to support training and optimization of software engineering agents

Research Intern supervised by Prof. Darko Marinov Sep. 2025-Jul. 2026

A large-scale empirical study of real-world software bug reports created by researchers, focusing on the traceability and verifiability of bug claims in academic papers.

- **Bug Report Dataset Construction** Built and curated a large-scale dataset linking bug claims in research papers to real-world bug reports; collected and standardized report-level metadata across multiple bug-tracking platforms.

- **Bug Report Recovery & Validation** Investigated research papers, artifacts, repositories, and authors’ public activity to recover supporting bug reports; manually validated report provenance, relevance, bug counts, and developer resolutions.
- **Empirical Analysis of Research Bug Reporting** Analyzed the traceability and consistency of bug-reporting claims, identifying recurring issues such as missing reports, undisclosed rejected reports, duplicate counting, and mismatches between claimed and developer-assigned statuses.
- **Project Coordination & Research Leadership** Coordinated a multinational research team, led data collection and validation efforts, maintained annotation guidelines and methodology, and contributed to analysis, interpretation, and paper writing.

SAP Labs Korea

Data Scientist
Jun. 2023-Aug. 2025

Focused on building LLM-based systems for automated debugging, reasoning extraction, and developer-assistance tools. Designed research-driven pipelines for analyzing historical bug reports, extracting structured reasoning, and enhancing fault localization and solution prediction.

Key Research Projects

- **Reasoning Path: New Approach to Capture Historical Bug Resolution Processes** Extracted structured reasoning steps (observation → thought → action → evaluation) from historical HANA bug reports to improve root cause analysis and solution generation. Developed evaluation metrics for structural consistency and contextual reliability.
- **Semantic Search for Bug Report Retrieval** Designed chunking + hierarchical retrieval (tree-structured RAG) pipeline that significantly improved relevance and precision over keyword-based search.
- **LLM-based Bug Report Summarization** Built an LLM summarization module that distilled essential debugging signals while enforcing sensitive-information filtering.

Seoul National University - Social & Computer Network Lab

Data Scientist
Jan. 2015-Feb. 2017

Conducted research on wireless networking, communication interference, and recommender systems using large-scale mobility datasets.

Selected Research Contributions

- Published at **IMC’15** and **NSDI’16** through joint industry–academia projects with Samsung Electronics.
- Developed machine learning models for path management in MPTCP and large-scale location-based recommender systems.

RESEARCH GRANTS

Total funding secured: \$950,000 (Korean government + private companies)
I led ideation, proposal writing, and served as Project Director for execution.

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|---|--------------------------|
| • AI Multi-agent for Intelligent Video Highlight Extraction (2025) | Secured \$100,000 |
| • Chrome Extension for Watermarking, LLM-based Service (2024) | Secured \$100,000 |
| • Cloud Web Storage Platform (2024) | Secured \$90,000 |
| • Webtoon-to-Animation Production Tool (2023-2024) | Secured \$500,000 |
| • Data Processing (2022–2023) Korea Data Agency | Secured \$160,000 |

INDUSTRY EXPERIENCE (SELECTED, RESEARCH-RELEVANT)

SAP Labs Korea

Analytical Reporting Developer
Nov. 2017-Aug. 2025

Developed enterprise-scale CI/CD monitoring and development-quality analysis systems.

- Built automated dashboards, ETL pipelines, and migration frameworks from on-premise to cloud.
- Designed data models for tracking development velocity, regression patterns, and CI health.

PatternTech

Director / Project Manager
Jun. 2020-Jul. 2026

Led AI projects combining LLMs, vision models, and large-scale data processing.

- Built Chrome-based LLM tools (translation, summarization, document search).
- Managed architecture for AI-assisted animation production systems.

Korea Information Technology Corporation *Senior Researcher*

Feb. 2017-Nov 2017

Developed IoT devices and real-time monitoring systems for health and senior care.

AWARDS

Honors and Awards

- Ceremonial speaker for the SAP Academy for Engineering graduation 2023
- SAP Early talent award 2019

Academic Scholarship

- Scholarship for academic excellence in Seoul National University (1 semester) 2016